EXPERIMENT No. → 1

1) **Aim** : To determine the wavelength of He-Ne Laser using diffraction grating.

2) **Apparatus Required** : Plane transmission grating, laser source with stand, measuring tape and a screen.

3) **Formula Used** : The wavelength λ of a laser is calculated by the following formula

$$\boxed{\lambda = \frac{e x_1}{D}}$$

where,

e = grating element

x_1 = mean distance of 1st order laser spot

D = distance of screen from the grating

λ = wavelength

$$e = \frac{2.54}{N}$$

where, N = No. of lines per inch on grating

= 15000 lines/inch

{1 inch = 2.54 cm}

$$\therefore \quad e = \frac{2.54}{15000} = 1.693 \times 10^{-4} \text{ cm}$$

The condition for n^{th} maxima is

$$e \sin\theta_n = n\lambda$$

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$$\lambda = \frac{e \sin\theta_n}{n}$$

But $n=1$, for first order of spectrum and $\sin\theta_n \simeq \tan\theta_n = \frac{x_1}{D}$. Therefore above equation becomes,

$$\lambda = \frac{e x_1}{D}$$

4) Diagram :

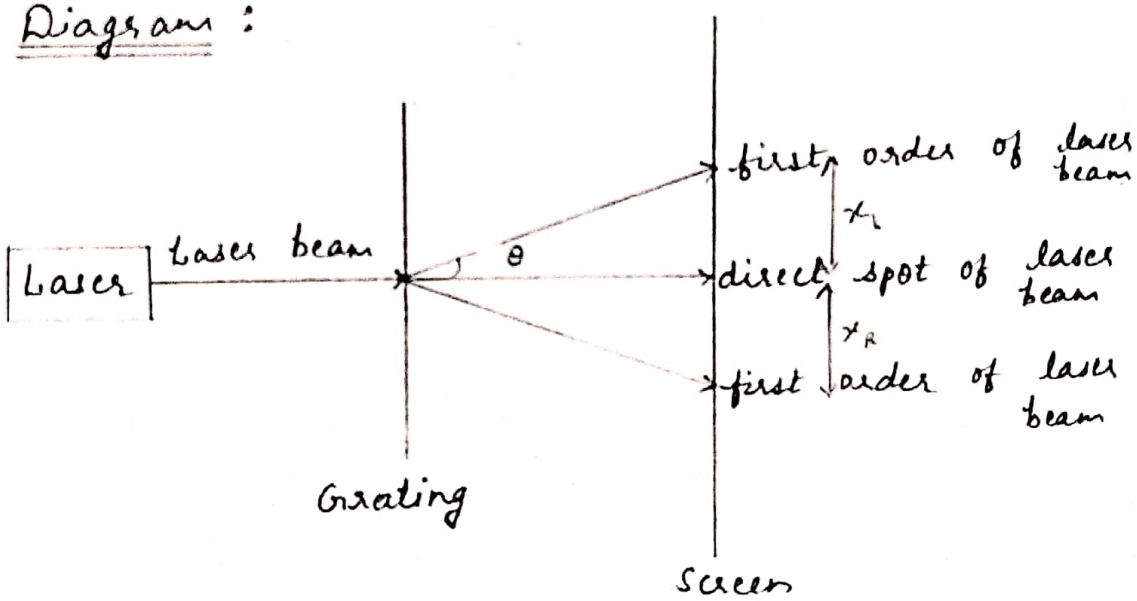

Fig: Experimental position of laser beam, grating and screen.

5) Observation Table :

S.No.	Distance of screen from grating. D (in cm)	Order of spectrum (n)	Distance from central maxima (in cm.)		Mean $x_1 = \frac{x_L + x_R}{2}$ (in cm.)
			x_L	x_R	
1.	52 cm	1	18.8	18.7	18.75
2.	66 cm	1	23	24	23.5
3.	100 cm	1	36	36	36

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6) Calculation :

I case :- $D = 52$ cm

$x_1 = 18.75$ cm

$e = 1.693 \times 10^{-4}$ cm

$$\lambda_1 = \frac{e x_1}{D} = \frac{1.693 \times 10^{-4} \times 18.75}{52}$$

$$= 0.61045 \times 10^{-4} \text{ cm}$$

$$= 6104 \text{ Å} \longrightarrow (1)$$

II Case :- $D = 66$ cm

$x_1 = 23.5$ cm

$e = 1.693 \times 10^{-4}$ cm

$$\lambda_2 = \frac{e x_1}{D} = \frac{1.693 \times 10^{-4} \times 23.5}{66}$$

$$= 0.6028 \times 10^{-4}$$

$$= 6028 \text{ Å} \longrightarrow (2)$$

III case :- $D = 100$ cm

$x_1 = 36$ cm

$e = 1.693 \times 10^{-4}$ cm

$$\lambda_3 = \frac{e x_1}{D} = \frac{1.693 \times 10^{-4} \times 36}{100}$$

$$= 0.6095 \times 10^{-4}$$

$$= 6095 \text{ Å} \longrightarrow (3)$$

Averaging eq^n ①, ② and ③ we get,

$$\lambda = \frac{\lambda_1 + \lambda_2 + \lambda_3}{3} = \frac{6104 + 6028 + 6095}{3} = 6075.6 \text{ Å}$$

$\lambda \cong 6076$ Å (calculated)

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7) **Percentage error:**

Standard wavelength = 6328 Å

Calculated wavelength = 6076 Å

$$\% \text{ error} = \frac{\text{Std. Value} - \text{Cal. Value}}{\text{Std. Value}} \times 100\%$$

$$= \frac{6328 - 6076}{6328} \times 100\%$$

$$= 3.98\%$$

8) **Result**: The wavelength of laser light is found to be 6076 Å. The percentage error is found to be 3.98%.

9) **Precautions**: (a) Use laser carefully and avoid to direct view.

(b) The grating ruled surface should be placed normally to the incident laser beam.

(c) The screen should be placed ⊥ to the incident laser beam.

(d) Grating should not be touched by hands.

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